

## Research interests

**General** formal methods, game theory, automata theory, logic, discrete mathematics

**Specific** reactive synthesis, verification, probabilistic systems, Markov decision processes

## Education

**PhD in Theoretical Computer Science** (ongoing, synopsis submitted)

*Tata Institute of Fundamental Research, Mumbai, India*

Advisor: Shibashis Guha

Thesis: *Window mean-payoff in turn-based stochastic games*

Relevant courses: Automata, verification, and infinite games; Descriptive complexity; Probability; Algebraic automata theory; Computational complexity; Algebra and computation; Online algorithms

**Bachelor of Science (Research) with major in Mathematics** (Aug 2016 – Jul 2020)

*Indian Institute of Science, Bangalore, India*

Relevant courses: Automata theory and computability; Introduction to scalable systems; Game theory; Graph theory; Combinatorics; Number theory; Measure theory

## Publications

- **Sure-almost-sure and Sure-limit-sure Window Mean Payoff in Markov Decision Processes** ([CONCUR 2026](#), [arXiv](#))

with Shibashis Guha

- **Expectation in Stochastic Games with Prefix-independent Objectives** ([CONCUR 2025](#), [arXiv](#))

with Laurent Doyen and Shibashis Guha

Presented at

- Seminar talk at MPI-SWS Kaiserslautern
- [CONCUR 2025](#)
- [Workshop on Automata and Games for Synthesis](#) (co-located with FSTTCS 2024)

- **Optimising Expectation with Guarantees for Window Mean Payoff in Markov Decision Processes** ([AAMAS 2025](#), [arXiv](#))

with Shibashis Guha

Presented at

- [Workshop on Automata and Games for Synthesis](#) (co-located with FSTTCS 2025)
- Seminar talk at Masaryk University, Brno
- Seminar talk at IST Austria
- [Highlights 2025](#)
- Seminar talk at IIT Bombay
- Formal Methods Update meeting 2025

- **Recognizing numbers** (ICLA 2025 (best student paper award), [arXiv](#))  
with Arnab Sur

Presented at

- [ICLA 2025](#)
- TCS Research Expo 2024
- STCS Student Symposium 2024

- **Stochastic Window Mean-payoff Games** (FoSSaCS 2024, LMCS)  
with Laurent Doyen and Shibashis Guha

Presented at

- [ACM ARCS 2025](#)
- Seminar talk at ENS Paris-Saclay
- [FoSSaCS 2024](#)
- [STCS Student Symposium 2023](#)
- [Formal Methods Update Meeting 2023](#)

## Other talks and presentations

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### Seminar talks at TIFR

- The Canadian traveller problem
- Sperner's lemma and the equidissection of regular polygons
- The connection between circuit complexity and first-order logic
- The complexity of solving simple stochastic games
- Courcelle's theorem
- Total-payoff games on graphs with windows
- Determinacy of two-player games with perfect information
- Vertex connectivity of Eulerian orientations

### Outreach sessions

- Games, Secrets, and Smart Thinking at [Chai and Why 2026](#)
- Sperner's lemma and its application to rent division at [Chai and Why 2025](#), [Vigyan Vidushi 2024](#)
- Voting mechanisms: Let's put it to a vote at [Chai and Why 2024](#)
- Beating Chinese Whispers and other communication problems at [Chai and Why 2023](#)
- Fun with graphs at [Chai and Why 2022](#)
- Impartial games at IISc Open Day 2020

## Experience

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### Reviewing

FMCAD 2026, MathOR, ATVA 2025, EC 2025, CAV 2025, CSL 2025, STACS 2025, LICS 2024

### Moderating

- Discussion on *Open problems and challenges in Reactive Synthesis* at [IndiCS Seminar on Automated Synthesis 2025](#)
- Panel Discussion on *Education and Careers in Logic (ECL)* at [ICLA 2025](#)

### Teaching assistance

Automata and Computability at TIFR (January – May 2025)

## Conferences attended

- 2025 • FSTTCS 2025 in BITS Pilani, Goa, India  
• IndiCS Seminar on Automated Synthesis 2025, Mysore, India  
• ATVA 2025 in IIIT Bangalore, India  
• Highlights 2025 in Saarland University, Saarbrücken, Germany  
• CONCUR 2025 in University of Aarhus, Denmark  
• Formal Methods Update meeting 2025 in DAU, Gandhinagar, India  
• ACM ARCS 2025 in PSG College of Technology, Coimbatore, India  
• ICLA 2025 in ISI Kolkata, India
- 2024 • FSTTCS 2024 in IIT Gandhinagar, India  
• Winter School on Verification 2024 in IIT Delhi, India  
• SAT 2024 in TCS Pune, India  
• ISLA 2024 in IIT Goa, India  
• ETAPS 2024 in Luxembourg (student volunteer)
- 2023 • FSTTCS 2023 in IIIT Hyderabad, India  
• Formal Methods Update meeting 2023 in IIT Goa, India
- 2022 • FSTTCS 2022 in IIT Madras, Chennai, India  
• FLoC 2022 in Technion, Haifa, Israel (student volunteer)

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Up to date as of 18 June 2026

Most recent version available at [pranshugaba.com/cv](https://pranshugaba.com/cv)

## Referees

**Shibashis Guha**

Reader

School of Technology and Computer Science,  
Tata Institute of Fundamental Research,  
Mumbai, India

[shibashis@tifr.res.in](mailto:shibashis@tifr.res.in)

**Laurent Doyen**

CNRS Researcher

Laboratoire Méthodes Formelles (LMF),  
ENS Paris-Saclay,  
Gif-sur-Yvette, France

[ldoyen@lmf.cnrs.fr](mailto:ldoyen@lmf.cnrs.fr)